

Mark schemes

Q1.

(a) (i) $r = \frac{294}{420} = 0.7$

AG. Accept any valid justification to the given answer

B1

1

(ii) $\{S_{\infty}\} \Rightarrow \frac{a}{1-r} = \frac{420}{1-0.7}$
 $\frac{a}{1-r}$ used

M1

$\{S_{\infty}\} = 1400$

1400 NMS mark as 2 / 2 or 0 / 2

A1

2

(iii) n th term $= 600 \times (0.7)^n$

If not B2 award B1 for $420 \times (0.7)^{n-1}$ OE

B2

2

(b) (i) $\{u_n\} = 248 - 8n$

Accept ACF

B1

1

(ii) $u_k = 0 \Rightarrow 8k = 248$

$248 - 8k = 0$ OE e.g. $240 + (k-1)(-8) = 0$
 ft if no recovery, on c's (b)(i) answer

M1

$k = 31$

A1

$\sum_{n=1}^k u_n = 240 + 232 + \dots + 0 = \frac{k}{2} [240 + 0]$

For $\frac{k}{2} [240 + 0]$ or for $\frac{k}{2} [c's u_1 + 0]$

OE e.g. $\frac{k}{2} [2 \times c's u_1 + (k-1)(-8)]$

M1

$$\sum_{n=1}^k u_n \quad (= 15.5 \times 240) = 3720$$

A1

4

[10]

Q2.

(a) $\{S_{25} = \frac{25}{2} [2a + (25-1)d]$
 $\frac{25}{2} [2a + (25-1)d]$ OE

M1

$$\frac{25}{2} [2a + 24d] = 3500$$

$$25(2a + 24d) = 7000 \text{ or } \left[\frac{50a + 600d}{2} = 3500 \right]$$

Forming equation and attempt to remove fraction or to expand brackets or better

M1

$$50a + 600d = 7000 \text{ (or better) so } a + 12d = 140$$

CSO AG Be convinced.

A1

3

(b) $5^{\text{th}} \text{ term} = a + 4d$
 $a + (5-1)d$ used correctly

M1

$$a + 12d = 140, a + 4d = 100 \Rightarrow 8d = 40$$

Solving $a + 12d = 140$ simultaneously with either $a + 4d = 100$ or $a + 5d = 100$ as far as eliminating either a or d .

M1

$$\Rightarrow d = 5$$

A1

$$\Rightarrow a = 80$$

A1

4

(c)
$$33 \left(3500 - \sum_{n=1}^k u_n \right) = 67 \sum_{n=1}^k u_n$$

Recognition that $\sum_{n=1}^{25} u_n = 3500$

M1

$$33 \times 3500 = 67 \sum_{n=1}^k u_n + 33 \sum_{n=1}^k u_n$$

Correct rearrangement PI

M1

$$100 \times \sum_{n=1}^k u_n = 33 \times 3500 \Rightarrow \sum_{n=1}^k u_n = 1155$$

A1

3

[10]

Q3.

(a)
$$\{S_{31}\} = \frac{31}{2} [2a + (31-1)d]$$

M1

$$31(a + 15d) = 310$$

Forming eqn and eliminating fraction or bracket

m1

$$a + 15d = 310/31; \quad a + 15d = 10$$

AG Completion to printed answer

A1

3

(b)
$$a + (21-1)d = 2[a + (16-1)d]$$

$a + (n-1)d$ used for at least one term

M1

$$\Rightarrow a = -10d; \Rightarrow -10d + 15d = 10$$

*Solving $a + 15d = 10$ simultaneously
 with an eqn in a and d obtained from
 $a + 20d = k[a + 15d]$ with $k = 2$ or with $k = 1/2$*

m1

$$d = 2$$

A1

3

(c) $u_1 = a = -20$

*ft on c's value for d in $a + 15d = 10$ or in
 another correct (dep on m1) equation in
 a and d
 The value for a must appear within c's soln for (c)*

B1F

$$\sum_{n=1}^k u_n = S_n = \frac{k}{2}[2a + (k-1)d]$$

*Condone n for k in M1 and A1F lines
 provided n replaced by k at a later stage*

M1

$$\frac{k}{2}[-40 + 2k - 2] = 0$$

*'= 0' can be implied by later line; ft on c's
 non-zero values for a and d*

A1F

$$k = 21$$

*Condone presence of $k = 0$
SC NMS $k = 21$ and with $d = 2$ found
 earlier award B2. If $k = 21$ but never see
 $d = 2$, award 0/4*

A1

4

[10]

Q4.

(a) (i) $\{S_{\infty}\} = \frac{a}{1-r} = \frac{10}{1-r}$

$\frac{a}{1-r}$ used

M1

$$\frac{10}{1-r} = 50 \text{ so } 1-r = \frac{10}{50} \Rightarrow r = \frac{4}{5}$$

AG Condone verification with the correct final statement but be convinced.

A1

2

(ii) 2nd term = ar

ar stated or used for the 2nd term. PI by ans'8'

M1

= 8

A1

2

(b) (i) 4th term = $a + 3d$; 8th term = $a + 7d$

Uses $a + (n - 1)d$ correctly at least once

M1

$$a + 3d = 10, \quad a + 7d = 8$$

*Both eqns. correct ft on c's (a)(ii) OE eg
8 = 10 + 4d*

A1F

$$\Rightarrow 4d = -2 \Rightarrow d = -0.5$$

OE fraction.

A1

3

(ii) $a + 3(-0.5) = 10$

*An appreciation that a is required in
(b)(ii) and a valid method to find a
anywhere or PI if $a = 11.5$ seen/used*

M1

$$\Rightarrow a = 11.5$$

A1F

Ft on c's non-zero value for d
ie using $a = 10 - 3d$ or $a = c's\ 8 - 7d$.
[c's 8 is candidate's answer to (a)(ii)]

$$\sum_{n=1}^{40} u_n = S_{40} = \frac{40}{2} [2a + (40 - 1)d]$$

$$\frac{40}{2} [2a + (40 - 1)d] \quad OE$$

M1

$$= 70$$

A1

4

[11]

Q5.

(a) (i) $ar = 375; ar^4 = 81$

For either OE or PI by next line

B1

$$\Rightarrow 375r^3 = 81$$

Elimination of a OE

M1

$$r^3 = \frac{81}{375} = \frac{27}{125} = 0.216 \Rightarrow r = 0.6$$

CSO AG Full valid completion
SC: Clear explicit verification, with
statement max B1 out of 3. (If considers
uniqueness then 3 is possible)

A1

3

(ii) $0.6a = 375$

OE; PI

M1

$$a = 625$$

A1

2

(b) $\frac{a}{1-r} = \frac{a}{1-0.6}$

$\frac{a}{1-r}$ **used** with $| \text{value of } r | < 1$

M1

$$S_{\infty} = \frac{625}{0.4} = 1562.5$$

ft on cand's value for a.... ie $2.5 \times a$

A1F

2

(c) $\sum_{n=6}^{\infty} u_n = \sum_{n=1}^{\infty} u_n - \sum_{n=1}^5 u_n$

M1

$u_3 = 0.6u_2 (= 225)$ and $u_4 = 0.6^2u_2 (= 135)$

Valid method to either find u_3 and u_4 or

use of $\{S_n\} \frac{a(1-r^n)}{1-r}$ for either $n = 5$ or $n = 6$

M1

$$\sum_{n=1}^5 u_n = 625 + 375 + 225 + 135 + 81 (= 1441)$$

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A1

$$\sum_{n=6}^{\infty} u_n = 1562.5 - 1441 = 121.5$$

A1

4

Alternative:

Recognise that the sum to infinity with first term u_6 is required

(M1)

Valid method to find $u_6 (= 0.6 u_5)$

(M1)

$$\sum_{n=6}^{\infty} u_n = \frac{81 \times 0.6}{1 - 0.6}$$

(A1)

$$= 121.5$$

(A1)

[11]

Q6.

(a) $u_1 = 12$

B1

$$u_2 = 3 \times 4^2 = 48$$

CSO AG (be convinced)

B1

2

(b) $r = 4$

B1

1

(c) (i) $\{S_{12}\} = \frac{a(1-r^{12})}{1-r}$

OE Using a correct formula with $n = 12$

M1

$$= \frac{12(1-4^{12})}{1-4}$$

Ft on answer for u_1 in (a) and r in (b)

A1ft

$$= \frac{12(1-4^{12})}{-3} = -4(1-4^{12}) = 4^{13} - 4$$

CAO Accept $k = 13$ for 4^{13} term

A1

3

(ii) $\sum_{n=2}^{12} u_n = (4^{13} - 4) - u_1$

$$= 67108848$$

B1

1

[7]

Q7.

(a) (i) $a + 19d = 181$;

$$a + 4d = 46$$

$$a + (n-1)d \text{ used; PI}$$

M1

$$\Rightarrow 15d = 181 - 46$$

A1

$$\Rightarrow d = 9$$

AG (be convinced)

A1

3

(ii) $a = 10$

B1

1

(b) $S_{20} = \frac{20}{2} [2a + (20-1)d]$

OE

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M1

$$\dots = 1910$$

A1

2

(c) $\sum_{n=1}^{50} u_n - \sum_{n=1}^{20} u_n$

OE

M1

$$\dots = 11525 - "1910" = 9615$$

ft on 11525 – c's S_{20}

A1ft



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